

Which Hand Gestures Make a Good Input Vocabulary?

Choosing gestures for AR glasses that are natural, rare in everyday life, distinct and meaningful

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Study 2 · results generated 2026-10-07

Abstract

A companion study ranked hand gestures by how natural they are for the hand [14]. Natural is not enough for input: the gestures the hand makes most easily are the ones it makes all day, so they fire by accident, and a vocabulary of individually good gestures can still contain lookalikes and gestures that begin like others. We treat vocabulary design as an assignment problem over 231 candidates (97 single gestures, including 16 new one-way thumb strokes, plus repeats, sequences and clutch chords) and four pressures: study 1's effort, everyday false triggers, clashes between gestures, and meaning. False triggers are counted the way MAGIC's Everyday Gesture Library counts them [3]: every gesture is slid along 1.32 h of everyday hand kinematics from 22 people [10] by a recognizer that absorbs speed and size and judges each body part against its own tolerance. Clashes (lookalikes, gestures hidden inside others, and prefixes that force a recognizer to wait) are computed for all 53,130 ordered pairs. Cheap gestures are common ones: study 1's effort and everyday false triggers are negatively rank-correlated ($\rho = -0.40$; $\rho = -0.48$ for laboratory effort alone, which shares no data with the everyday stream). A vocabulary chosen for naturalness alone would fire 60.29 times an hour of everyday use and give the wrong command on 70% of uses. The optimiser, run under 200 joint uncertainty draws, recommends a set built entirely on the thumb against the side and pad of the index and middle finger (taps, holds and one-way strokes), close to Meta's microgestures and Neural Band; it fires 0.00 times an hour within the resolution of the data. Of 6 independent registered predictions 4 pass; the method changed after development runs, and every change is disclosed. Everything is model output; no person has performed these vocabularies.

1 Introduction

Study 1 asked which gestures are most natural for the hand on its own, before any sensor or meaning [14]. Its answer—small wrist and forearm movements, partial opening of the hand, the free thumb, and the thumb on the index—is sound for comfort and wrong for input. An input gesture must be easy to perform *and* unlike anything the hand does without meaning to. This is the Midas touch problem [9]: when the act that controls a system is also an ordinary act, the system cannot tell intent from activity. For gaze it is famous; for the hand it returns as soon as small, everyday-like movements issue commands [1].

An input vocabulary also has properties no single gesture has. Two gestures can be individually excellent and indistinguishable from each other. A short gesture can be the opening of a longer one, so a recognizer must wait before acting on the short one (the double-click delay). And commands come in families—scroll up and down, next and previous—that are easier to learn when their gestures mirror each other [21]. Study 1 is a dictionary of costs; this study chooses a language from it.

Contributions.

1. **An everyday false-trigger measure for hand gestures in joint angles:** MAGIC’s method [3, 13] applied to 1.32 h of everyday hand kinematics, with a recognizer that absorbs speed and size and judges each body part separately.
2. **Clash measures for a whole vocabulary:** lookalikes, gestures hidden inside others, and prefix latency, computed for every ordered pair of 231 candidates (the continuous counterpart of Proton’s static conflict check [12]).
3. **A vocabulary optimiser** over effort, false triggers, clashes, meaning and structure, with no fitted weights, robust selection across 200 joint uncertainty draws, ablations, shipping baselines and four sensor lenses.
4. **Registered predictions**, written before any output, and a full account of the method changes made after development runs.

2 Related work

Everyday false triggers. MAGIC records unlabelled sensor data from people living ordinary days (an Everyday Gesture Library, EGL) and counts how often a candidate gesture would have fired [3]. Designers without the EGL produced gestures that fired 52.1 times per gesture per hour; with it, 1.9; a 5.3 h slice predicted the rest of a 58 h library ($r = 0.87$). MAGIC Summoning indexes the EGL to suggest gestures that rarely occur [13]. DoubleFlip, a fast double wrist flip used as a delimiter, fired less than once per eight hours across 2,169 h of phone motion and removed the false positives of a shake gesture it preceded [22]. SoloFinger exploits the rarity of lone-finger movement during grasping [23]. Gesture spotting treats “is this a gesture at all?” as recognition in its own right [15].

Conflicts within a set. Proton writes multitouch gestures as regular expressions and checks a set for conflicts, including prefixes [12]; Quill warns pen-gesture designers about pairs a recognizer or a person would confuse [17].

Vocabulary design as optimisation. Stern, Wachs and Edan combine intuitiveness, comfort and recognition accuracy into a multi-criteria gesture-to-command assignment and display its Pareto front [4, 24]. Li et al. had 40 users link 33 ergonomist-designed microgestures to 20 AR/VR commands, mainly by preference [16]. We follow the multi-criteria framing; what is new is that comfort comes from a computed body model and recognition from computed clashes and everyday false triggers rather than a trained classifier.

Meaning. Elicitation agreement for microgestures is low (mean AR 0.19, maximum 0.65 in Chan et al. [5]; [25, 27]), and elicited gestures copy prior technology [20]. Recall failures are mostly failures to remember which gesture goes with which command [21], which motivates rewarding consistent, mirrored mappings.

Shipping vocabularies. visionOS pairs gaze targeting with a thumb–index tap, double tap, pinch-and-hold and pinch-and-drag [2]. Quest microgestures are a thumb tap on the side of the index and four thumb swipes along and across it [18]. Meta’s Neural Band (EMG) uses index and middle pinches, thumb slides along the index and pinch + wrist twist [11, 19].

3 Method

3.1 Commands

Two conditions. *Gaze + hand*: gaze picks the target and the hand gives the command, so the vocabulary needs only commands that do not act on a visible button: select, inspect, back, menu, dismiss, and four bidirectional controls (scroll, page, value, zoom), 13 slots. *Hand only*: the hand also moves focus horizontally and vertically, 17 slots. Each slot has a frequency of use and a consequence when it fires by mistake; both are assumptions (Appendix C) and are perturbed in the uncertainty draws.

3.2 Candidates

All of study 1’s reachable canonical gestures, plus 16 one-way thumb strokes: study 1’s slides go out and back, so they have no direction, and a vocabulary needs a gesture and its mirror image. Each stroke reuses a study 1 slide (same approach, contact envelope, finger and segment) with the thumb travelling one way, along the finger or across the side of the index. Combinations: repeats of every tap and of the main swings (27); the cheapest fluent sequences within a core of cheap gestures and contacts, at most three per first gesture (40); and clutch chords, a held contact or handshape plus any compatible movement (67). Every candidate is scored by study 1’s cost model unchanged.

3.3 Features

Each trajectory is resampled to 25 Hz over the span where it leaves HOME and expressed in 29 features in units of everyday standard deviation: the 17 everyday channels, four finger DIP joints and forearm rotation (scaled from published ranges), contact weight on each finger, contact position and pressure. Templates are gestures *as performed*: a moving finger drags its neighbours by Wilhelm et al.’s enslaving fractions [26].

3.4 The recognizer

A template is compared with a window of movement (everyday life, or another gesture) by a recognizer with three tolerances. *Speed*: the window may last $0.67\text{--}1.5\times$ the template (uniform time scaling [6]). *Size*: the template’s movement is scaled by one fitted $\alpha \in [0.5, 1.67]$, the ratio of study 1’s size classes. *Parts*: features are grouped by body part (thumb, each finger, spread, wrist, forearm, contact, pressure). For each part the residual movement (plus a small posture term, $\beta = 0.1$, on moving parts) is compared with that part’s own tolerance:

$$\tau_g = \sqrt{f_g^2 + (c s_g)^2} \text{ for parts the gesture moves,} \quad \tau_g = \sqrt{f_g^2 + v^2} \text{ for the rest,} \quad (1)$$

where f_g is everyday stillness (the 10th percentile of 1 s movement of that part), s_g the part’s movement in the template, c the recognizer’s forgiveness (signal-dependent motor noise [7]) and $v = 0.75\text{ SD}$ the drift allowed in parts the gesture does not mean to move. The distance is the worst part relative to its tolerance; 1 is the edge of recognition. Taking the worst part means a wrist bend added to a held pinch is a different gesture even though the pinch dominates the movement. Parts that only follow by coupling count as drift.

The forgiveness c is the study’s key assumption. We set $c = 0.3$ and sweep it from 0.15 to 0.6. As context, a recognizer that had to accept every near-optimal posture the contact solver finds for the same contact would need $c \approx 1.5$; those alternatives span every acceptable posture, not how one person repeats a contact, so this is an upper bound.

3.5 Everyday false triggers

KIN-MUS UJI [10]: 22 right-handed adults performing 26 activities of daily living, 18 calibrated angles at 100 Hz. The three phases of each activity are contiguous and were joined; frames outside physiological wrist limits were removed (study 1's artefact filter), runs shorter than 2 s dropped, leaving 649 runs and 1.32 h (84% of raw frames) at 25 Hz. Every template slides along every run; a hit is a window within tolerance that is the best within half a template length on either side, so one movement counts once. False triggers per hour (FT) are hits divided by hours. Contacts are matched on their kinematics (a thumb that reaches the index pad is where it would be when touching it). Forearm rotation, DIP joints and pressure are not in the data; a gesture whose movement is mostly made of them gets the median FT of its study 1 family, and a combination is capped by its rarest part. With 1.32 h of data, rates below about one an hour cannot be resolved.

3.6 Clashes

For each ordered pair (template A , performance B): the whole-against-whole distance; the best embedding of A anywhere inside B padded with rest; the best embedding at B 's opening; how far B had moved before an inside embedding; and how long B takes to move on after a prefix match. The recognizer is lazy with lookback: it waits while a longer gesture remains possible. So $P(\text{recognise } B \mid \text{perform } A)$ is the larger of the whole-distance match and an inside embedding of B in A preceded by rest-like movement; a match at the opening is not a misfire but adds latency to the shorter gesture. Recognition has a soft edge (certain below 0.9 of tolerance, impossible above 1.1).

3.7 Meaning and structure

Priors on the elicitation agreement scale [25]: shipping conventions up to 0.6, elicited mappings 0.2–0.3 (mostly Chan et al. [5]), physical metaphors at most 0.24, anything else 0.02. Structure rewards mirrored bidirectional controls (exact mirror 1, same form 0.5) and command groups (act, system, navigate, adjust) that share one form.

3.8 Objective and search

Five components, each 0 (best) to 1: frequency-weighted study 1 effort; $1 - e^{-H/H_0}$ with H the consequence-weighted false triggers per hour; $1 - e^{-(K/K_0 + L/L_0)}$ with K expected wrong commands per use and L expected waiting per use; 1– meaning relative to the best available; and structure. The objective is their mean. No study ranks the components, so weights are equal by default, the front across weightings is reported, and robustness draws sample them. The assignment is a quadratic assignment problem, searched by simulated annealing from greedy and random starts with exhaustive local descent.

Robust selection. Each of 200 joint draws samples the template configuration (posture weight $\beta \in \{0, 0.1, 0.3\}$, coupling on or off), the forgiveness ($c \times \{0.75, 1, 1.5\}$), a subject bootstrap of the everyday data, frequencies and consequences (log-normal, $\times/\div$ up to 2), the scales H_0, K_0, L_0 and the weights (Dirichlet, concentration 5). Every draw's optimum joins a pool of candidate vocabularies. The objective is not comparable across draws (each has its own weights and scales), so each candidate is ranked within every draw and the recommended vocabulary is the one with the best mean rank (a Borda count).

3.9 Sensor lenses

Headset cameras, an EMG wristband, a thumb ring and an index ring, each a per-feature sensitivity from the sensing review behind study 1 (archived v1 priors), applied as the share of each feature in the squared distance; EMG also mixes ring and little 80% towards their mean [8]. Everyday matching and clashes are recomputed under each lens with the body’s tolerances, and the recommended vocabulary is reselected.

3.10 Registered predictions

Fifteen claims were written before any study 2 output and hashed (Appendix D): six independent (from sources that are not inputs to the measure they test), four convergent and five structural expectations about the method itself.

4 Results

4.1 Natural gestures misfire

Figure 1 places every single gesture by effort and everyday false triggers. Cheap gestures are common: $\rho = -0.40$ between study 1’s composite and FT, and $\rho = -0.48$ for its laboratory terms alone ($n = 77$), which share no data with the everyday stream. Wrist bends, study 1’s most natural gestures, have a median of 11 false triggers an hour; lone finger movements 1.5; thumb contacts 0 and the new strokes 0 (Table 1). The rarity of contacts holds until the recognizer becomes very forgiving (Figure 2).

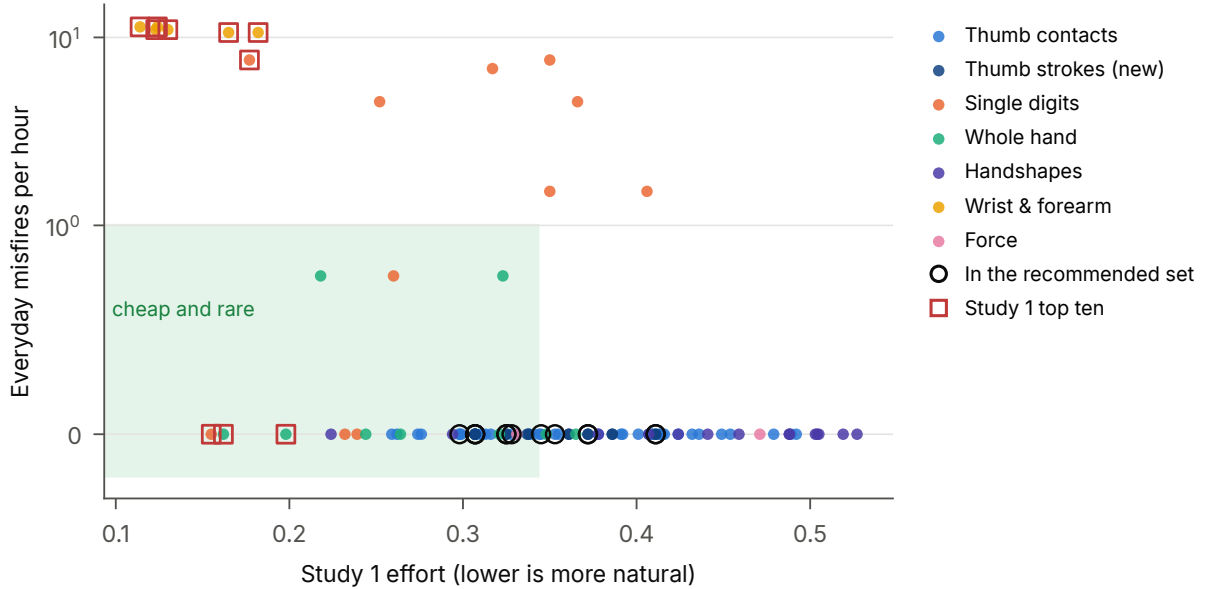


Figure 1: Every single gesture by study 1 effort and everyday false triggers per hour (symmetric log scale; *model output*). Squares: study 1’s ten most natural gestures. Rings: the recommended gaze + hand vocabulary.

4.2 The recommended vocabularies

Tables 2 and 3 give the recommended sets. The gaze + hand set fires 0.00 times an hour of everyday use and gives a wrong command on 0% of uses; the hand-only set 0.00 and 0%. Neither set needs a combination (0 and 0 slots): enough distinct thumb contacts and strokes exist that repeats, sequences and clutch chords never pay their extra effort. Stability is low at the level

Table 1: Median everyday false triggers per hour by family at $c = 0.3$ (*model output*; gestures the everyday data cannot see are excluded).

Family	n	Median FT/h
Thumb contacts	39	0
Single digits	11	1.5
Whole hand	9	0
Handshapes	14	0
Wrist & forearm	6	11
Force	2	0
Thumb strokes (new)	16	0

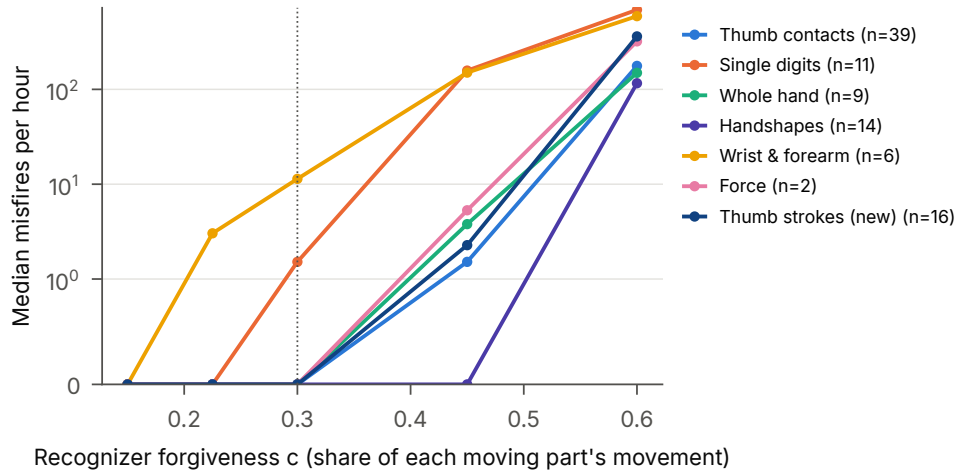


Figure 2: Median false triggers per hour by family as the recognizer's forgiveness c grows (*model output*). The dotted line is the default.

of exact gestures and higher at the level of form: many slots have near-equivalent alternatives (another segment of the same finger, the other side of the index), and 193 distinct optima arose across draws.

Table 2: Recommended gaze + hand vocabulary (*model output*). FT/h: everyday false triggers per hour. Exact: share of uncertainty draws whose optimum keeps this gesture; Form: keeps its form. Names link to the demonstration.

Command	Gesture	FT/h	Meaning	Exact	Form
Select	Index middle side tap	0	convention	89%	100%
Inspect / more	Index pinch hold	0	convention	36%	100%
Back	Middle pinch	0	convention	69%	76%
Menu / home	Middle middle pad press-hold	0	–	1%	53%
Dismiss / close	Middle distal pad tap	0	–	24%	54%
Scroll down	Index distal side stroke out	0	convention	43%	89%
Scroll up	Index distal side stroke in	0	convention	44%	89%
Next	Index distal side roll up	0	convention	34%	86%
Previous	Index distal side roll down	0	convention	34%	86%
Increase	Middle distal pad stroke in	0	elicitation	5%	23%
Decrease	Middle distal pad stroke out	0	elicitation	5%	23%
Zoom in	Middle middle pad stroke in	0	elicitation	4%	22%
Zoom out	Middle middle pad stroke out	0	elicitation	4%	22%

Table 3: Recommended hand-only vocabulary (*model output*).

Command	Gesture	FT/h	Meaning	Exact	Form
Select	Index pinch	0	convention	23%	100%
Inspect / more	Index distal pad press-hold	0	elicitation	7%	100%
Back	Middle pinch	0	convention	64%	67%
Menu / home	Middle distal pad press-hold	0	–	6%	59%
Dismiss / close	Middle middle pad press-hold	0	–	1%	59%
Scroll down	Index distal side stroke out	0	convention	25%	87%
Scroll up	Index distal side stroke in	0	convention	25%	87%
Next	Index middle side roll up	0	convention	16%	83%
Previous	Index middle side roll down	0	convention	16%	83%
Increase	Middle middle pad stroke out	0	elicitation	6%	21%
Decrease	Middle middle pad stroke in	0	elicitation	6%	21%
Zoom in	Middle distal pad stroke in	0	elicitation	7%	20%
Zoom out	Middle distal pad stroke out	0	elicitation	7%	20%
Focus right	Index distal side roll up	0	convention	14%	98%
Focus left	Index distal side roll down	0	convention	14%	98%
Focus down	Index middle side stroke out	0	convention	19%	94%
Focus up	Index middle side stroke in	0	convention	19%	94%

4.3 What drives the choice

Switching off one consideration at a time (Table 4, Figure 4) shows the division of labour. Choosing for naturalness alone puts most commands on the wrist; that set fires 60.29 times an hour and gives the wrong command on 70% of uses (hand only: 60.29 and 49%). Without meaning, 13 of 13 slots change: safety and effort alone do not single out the conventions, so their convergence with shipping systems is partly carried by the meaning priors, which encode those conventions (legacy bias [20]).

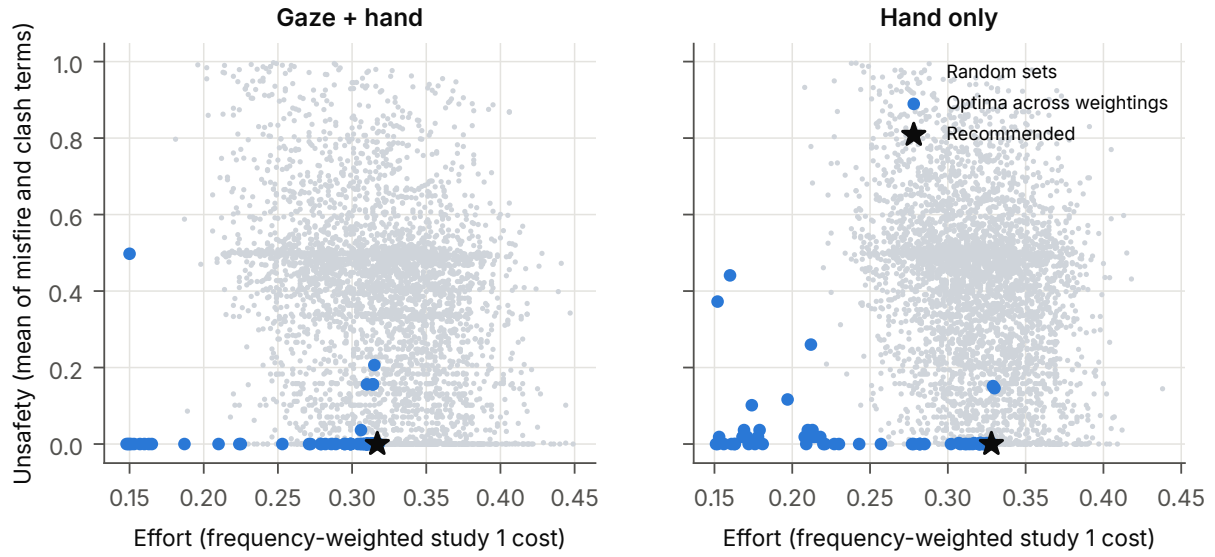


Figure 3: Effort against unsafety for random vocabularies, the optima across weightings and the recommended set (*model output*).

Table 4: Ablations: the default optimum with one consideration removed (*model output*). Changed: slots that differ from the default optimum; FT/h and Wrong for the whole set.

Variant	Gaze + hand			Hand only	
	Changed	FT/h	Wrong	Changed	Wrong
Without meaning	13/13	0.00	0%	17/17	0%
Body only (no meaning, no structure)	13/13	0.00	0%	17/17	0%
Without everyday false triggers	7/13	0.00	0%	10/17	0%
Without clashes	6/13	0.00	1%	14/17	1%
Naturalness only (study 1 cost)	13/13	60.29	70%	17/17	49%
Naturalness + meaning (no safety)	6/13	0.00	0%	16/17	1%

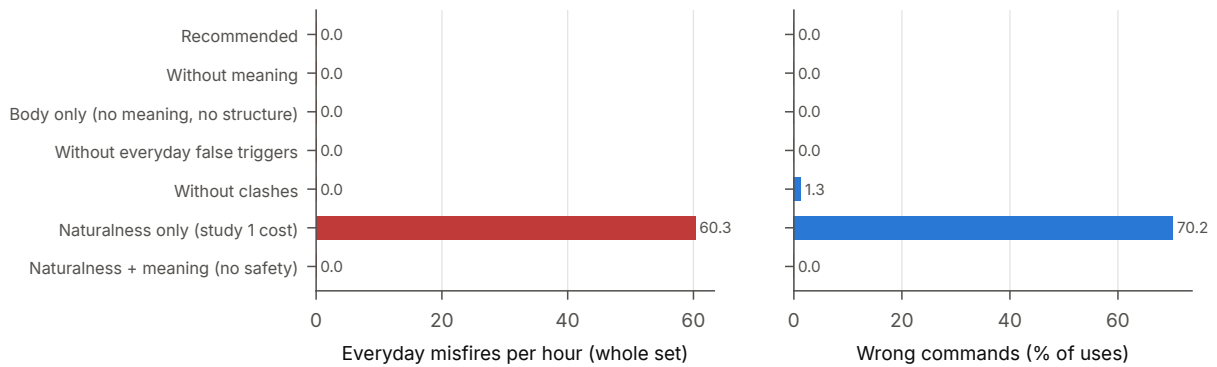


Figure 4: Everyday misfires and wrong commands of the recommended set and of the ablations (gaze + hand; *model output*).

4.4 Clashes

At the default tolerance 314 ordered pairs clash (24 among single gestures). The largest clashes are between gestures that share a contact and differ in a small part, and between holds and the movements made while holding (Table 5, Figure 5). A held pinch recognizer that tolerates drift elsewhere also accepts a pinch with a wrist bend, which is the ambiguity visionOS resolves with a movement threshold before a hold counts.

Table 5: Strongest clashes among single gestures (*model output*). *P*: probability the second is recognised when the first is performed.

Performed	Recognised	<i>P</i>	Wait (ms)
Close hand	S handshape	1.00	0
Open hand	Index lift	1.00	0
Open hand	Middle lift	1.00	0
Open hand	Ring lift	1.00	0
Open hand	Little lift	1.00	0
Ring pinch	Ring pinch hold	0.80	0
Little distal pad tap	Little distal pad press-hold	0.76	0
Middle pinch	Middle pinch hold	0.73	0
Little middle pad tap	Little middle pad press-hold	0.71	0
Index pinch	Index pinch hold	0.67	0
Wrist flex 30°	Wrist extend 30°	0.66	0
Wrist extend 30°	Wrist flex 30°	0.66	0
Little pinch	Little pinch hold	0.66	0
Index distal pad tap	Index distal pad press-hold	0.65	0

4.5 Against shipping systems

Each baseline is scored on its own commands next to the optimum for the same commands (§A). visionOS scores 0.289 against 0.062: in this body-only space its pinch-and-drag controls are pinch + wrist chords, which clash with its own pinch-and-hold (wrong command on 25% of uses), a clash the real system resolves with arm movement and a movement threshold. Quest’s microgestures (0.073 vs 0.061) and the Neural Band (0.077 vs 0.066) are close to the optimum.

4.6 Through a sensor

Under the lenses the recommended set changes in 8 (cameras), 6 (EMG), 12 (thumb ring) and 10 (index ring) of 13 gaze + hand slots (Figure 6, Table 9).

4.7 Registered predictions

4 of 6 independent, 3 of 4 convergent and 0 of 5 structural claims pass (Table 6). Table 7 shows how the false-trigger claims depend on the forgiveness *c*.

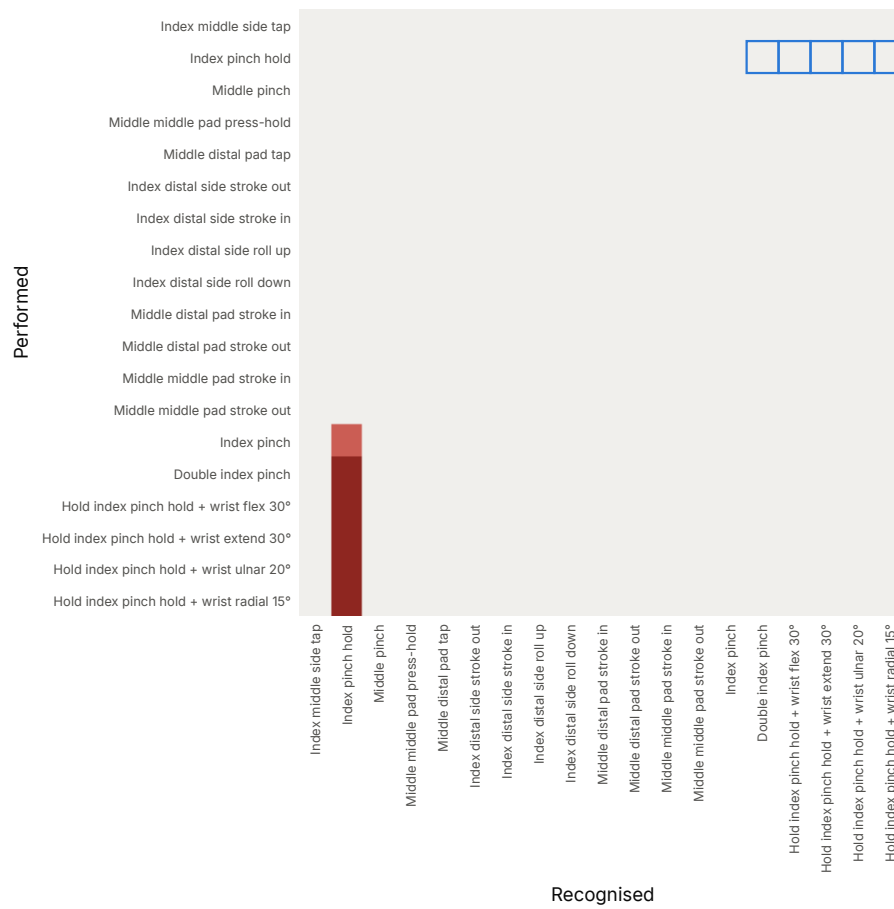


Figure 5: Confusion among the recommended gaze + hand set and the visionOS baseline (*model output*). Rows performed, columns recognised; blue outlines: prefix pairs that force a wait.

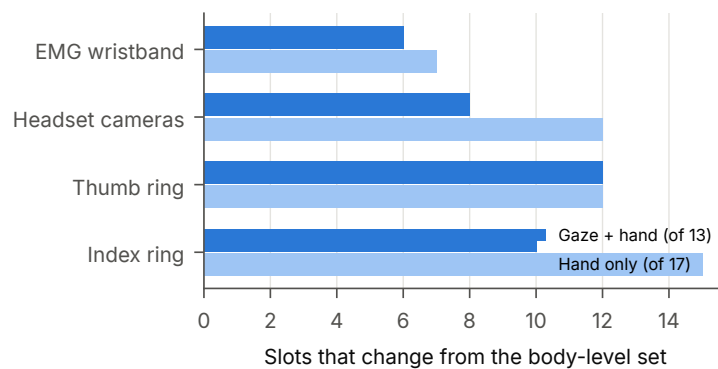


Figure 6: Slots of the recommended set that change under each sensor lens (*model output*).

Table 6: Registered predictions, evaluated by the final method (*model output* against claims written before any output).

Prediction (source)	Kind	Result
Lone-finger movements are rarer in everyday hand use than whole-hand movements: the median false-trigger rate of single-finger curls and lifts (index to little, medium) is below that of the whole-hand synergies (close, open, knuckle bend, hook, spread; medium). Sharma et al. 2021 (SoloFinger). lone fingers 4.55/h vs whole hand 0.00/h (medians)	independent	Fail
Thumb contacts on the index false-trigger less than wrist bends: the median rate over thumb→index contacts (tip, pad, side; tap, hold, slide) is below the median over wrist flex, extend, radial and ulnar (medium). Meta microgestures and Neural Band (design choice for always-available input). thumb→index 0.00/h vs wrist 11.00/h (medians, n = 14)	independent	Pass
Holding an index pinch as a clutch cuts the false triggers of wrist bends at least tenfold: the median over the four medium wrist bends of $(FT(wrist) + \epsilon) / (FT(pinch\ hold + wrist) + \epsilon)$ is ≥ 10 , with ϵ one event per data set. Ruiz & Li 2011 (DoubleFlip); Ashbrook & Starner 2010 (push-to-gesture). median ratio 15.50 (16.00, 16.00, 15.00, 15.00)	independent	Pass
The most natural gestures misfire often: the median false-trigger rate of study 1’s ten most natural observable gestures is at least 5 per hour. Ashbrook & Starner 2010 (designers without an EGL: 52 hits per gesture per hour; > 5/h counted as poor). median 4.17/h over wrist-extend-m, wrist-flex-m, thumb-abduct-m, open-m, wrist-ulnar-m, lift-index-m, wrist-radial-m, adduct, spread-m, shape-c	independent	Fail
Cheaper is commoner, on laboratory evidence alone: across canonical observable gestures, study 1’s laboratory-only cost is negatively rank-correlated with the false-trigger rate. MAGIC (comfortable designs collide with everyday motion); laboratory terms share no data with the everyday stream. $\rho = -0.48$ (n = 77)	independent	Pass
Every shipping baseline beats at least 90% of random vocabularies on its own commands under the equal-weight objective. visionOS, Meta microgestures, Meta Neural Band (expert-designed, shipping). visionos 100.0% · quest 100.0% · neural 100.0%	independent	Pass
Cheaper is commoner: across canonical observable gestures, study 1’s composite cost is negatively rank-correlated with the false-trigger rate. Study 1 everyday terms (same data as the false-trigger stream). $\rho = -0.40$ (n = 77)	convergent	Pass

Table 6: Registered predictions, evaluated by the final method (*model output* against claims written before any output).

Prediction (source)	Kind	Result
Smaller is commoner: for at least 80% of observable movement types, $FT(\text{small}) \geq FT(\text{medium}) \geq FT(\text{large})$. Method (smaller movements sit closer to stillness). 14/23 types monotone	convergent	Fail
Performed with their natural coupling, ring and little movements look alike: $d(\text{curl ring}, \text{curl little}) < d(\text{curl index}, \text{curl middle})$, and the same for lifts. Wilhelm et al. 2013 enslaving (the coupling leak input); Hu et al. 2015 (ring/little EMG covary). curl R-L 2.43 vs I-M 2.75; lift R-L 2.32 vs I-M 2.73 (τ units)	convergent	Pass
The gaze + hand flagship puts select on a thumb→index contact. visionOS, Meta, Chan 2016 (all put select on thumb-index; select’s meaning prior encodes it). select = side-tap-index-middle	convergent	Pass
Every tap and hold on the same contact site clash at the default tolerance (lookalike, embedding or prefix latency). Construction (a tap and a hold on the same site share their opening). 12/14 pairs clash	structural	Fail
Neighbouring segments of one finger look more alike than the same segment of neighbouring fingers: the median $d(\text{pad tap F distal}, \text{pad tap F middle})$ is below the median $d(\text{pad tap F distal}, \text{pad tap F+1 distal})$. Construction (the thumb travels less between segments than between fingers); cf. DigitSpace eyes-free accuracy. segments 6.30 vs fingers 4.98 (τ units)	structural	Fail
The hand-only flagship fills a larger share of its slots with combinations (repeats, sequences, chords) than the gaze + hand flagship. Method (more commands from the same gesture space). hand only 0.00 vs gaze + hand 0.00	structural	Fail
At least two thirds of the gaze + hand flagship’s slots keep their gesture in at least half of the joint uncertainty draws. Method (robustness). 2/13 slots $\geq 50\%$	structural	Fail
Of the sensor lenses, the headset camera changes the most gaze + hand flagship slots relative to the body-level flagship (strictly more than EMG and than either ring). v1 sensor sensitivities (cameras lose force and occluded contacts). emg 6 · camera 8 · thumb-ring 12 · index-ring 10	structural	Fail

4.8 Robustness

One-at-a-time changes and how many slots of the default optimum each moves are in Table 8. The vocabulary is most sensitive to the forgiveness c .

Table 7: False-trigger claims across the forgiveness c (*model output*).

Claim	$c = 0.15$	$c = 0.225$	$c = 0.3$	$c = 0.45$	$c = 0.6$
lone-fingers-rare	×	×	×	×	×
index-contacts-beat-wrist	×	✓	✓	✓	✓
clutch-tenfold	×	×	✓	✓	✓
natural-top-misfire	×	×	×	✓	✓
lab-cost-vs-ft	×	✓	✓	✓	✓
composite-cost-vs-ft	×	✓	✓	✓	✓
amplitude-monotone	✓	✓	×	×	×

Table 8: One-at-a-time sensitivity: slots of the default optimum that change (*model output*).

Change (one at a time)	Gaze + hand	Hand only
still parts v 0.5	6/13	8/17
still parts v 1.25	11/13	12/17
contact/site/pressure $\times 0.5$	8/13	12/17
contact/site/pressure $\times 2$	8/13	6/17
β 0, coupling on	10/13	16/17
β 0, coupling off	10/13	16/17
β 0.1, coupling off	11/13	12/17
β 0.3, coupling on	10/13	8/17
β 0.3, coupling off	9/13	12/17
noise 0.15	7/13	16/17
noise 0.225	7/13	10/17
noise 0.45	12/13	17/17
noise 0.6	8/13	13/17

5 Discussion

Natural and safe pull apart. The tension that made study 1’s answer feel wrong for input is real and measurable: the hand’s cheapest movements are its commonest. It does not depend on study 1’s everyday terms; laboratory effort alone shows it. The way out is not a better naturalness score but a selection step that charges for everyday resemblance and for clashes.

The thumb on the index is the input surface. Contacts and one-way strokes of the thumb on the side and pad of the index (and middle) are the cheapest gestures that everyday object handling almost never reproduces. That is also where every shipping system has converged. Part of the convergence is imported through the meaning priors; the ablation without meaning shows how much the body and the everyday data decide on their own.

Clutches. Holding a pinch while bending or turning the wrist turns the commonest movements into rare ones (median ratio 15.50 (16.00, 16.00, 15.00, 15.00)). That is the design logic of DoubleFlip’s delimiter [22], of MAGIC’s push-to-gesture anecdote [3] and of the Neural Band’s pinch + twist. The recommended sets do not need it: one-way thumb strokes on the middle finger cover the continuous controls more cheaply. Clutches are the way to keep wrist movements if a design wants them (the mean-objective selection, which weights draws with large objectives, chose pinch + wrist chords for scrolling and paging), and a hold recognizer must then be strict about drift, or it fires during every chord (Table 5).

6 Limitations

- **No people.** Recall, real false triggers, recognition and preference are unmeasured. A user study of the recommended sets is the next step.
- **The forgiveness is assumed.** No data on people repeating free-air gestures fixes c ; results are reported across a fourfold range and are most sensitive to it.
- **Short, narrow everyday data.** 1.32 h of goal-directed object handling, right hand only. No walking, talking with the hands, phone use or idle fidgeting, so communicative shapes are under-tested; MAGIC's EGL was 58 h of whole days. Rates below about one an hour are not resolved.
- **Unobservable movement.** Forearm rotation, finger tips and pressure are missing from the everyday data; their rates are estimates.
- **Idealised templates.** Gestures are study 1's trajectories with enslaving added; real performances vary in ways the tolerance only approximates.
- **Weak meaning priors** partly encode legacy conventions.
- **Lenses** are sensitivity priors, not sensor simulations.
- **No arm movement and one hand.** Pinch-and-drag and two-hand gestures are outside the space.

7 Conclusion

A vocabulary for AR glasses cannot be read off a naturalness ranking. Charging for everyday resemblance and for clashes moves the choice from the wrist to the thumb against the index and middle finger, and lands close to what shipping systems do; a held pinch is what makes wrist movements usable if a design wants them. The decisive next step is empirical: people performing the recommended sets through a day, to measure false triggers, recall and confusions directly.

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A Baselines

visionOS Tap, double tap, pinch and hold, pinch and drag. Zoom and rotate use two hands (outside this space); drags move the arm, approximated here by the wrist. Score 0.289 against 0.062 for the optimum on the same commands; it beats 100% of random sets.

Command	Shipping	Optimum
Select	Index pinch	Index middle side tap
Inspect / more	Index pinch hold	(same)
Menu / home	Double index pinch	Double middle pinch
Scroll down	Hold index pinch hold + wrist flex 30°	Index distal side stroke in
Scroll up	Hold index pinch hold + wrist extend 30°	Index distal side stroke out
Next	Hold index pinch hold + wrist ulnar 20°	Index distal side roll down
Previous	Hold index pinch hold + wrist radial 15°	Index distal side roll up
Wrong commands per use 0.245 vs 0.000; waiting 109 vs 0 ms per use; misfires 0.00 vs 0.00 per hour		

Quest microgestures Thumb tap on the middle segment of the index’s side, and four thumb swipes on it (forward, backward, left, right). Score 0.073 against 0.061 for the optimum on the same commands; it beats 100% of random sets.

Command	Shipping	Optimum
Select	Index middle side tap	(same)
Focus right	Index middle side roll down	Index distal side stroke out
Focus left	Index middle side roll up	Index distal side stroke in
Focus down	Index middle side stroke in	Index distal side roll down
Focus up	Index middle side stroke out	Index distal side roll up
Wrong commands per use 0.000 vs 0.000; waiting 11 vs 0 ms per use; misfires 0.00 vs 0.00 per hour		

Neural Band Index pinch acts, middle pinch goes back, a double middle pinch wakes the display, the thumb slides along the index to navigate, and pinch + wrist twist sets volume. Mappings are from press reports, not a published spec. Score 0.077 against 0.066 for the optimum on the same commands; it beats 100% of random sets.

Command	Shipping	Optimum
Select	Index pinch	Index middle side tap
Back	Middle pinch	(same)
Menu / home	Double middle pinch	Middle distal pad tap
Increase	Hold index pinch hold + fore-arm pronate 45°	(same)
Decrease	Hold index pinch hold + fore-arm supinate 45°	(same)
Focus right	Index distal side stroke out	Index distal side stroke in
Focus left	Index distal side stroke in	Index distal side stroke out
Focus down	Index distal side roll down	(same)
Focus up	Index distal side roll up	(same)
Wrong commands per use 0.000 vs 0.000; waiting 0 vs 0 ms per use; misfires 0.00 vs 0.00 per hour		

B Sensor lenses

Table 9: Recommended gaze + hand set under each lens (*model output*).

Command	EMG wristband	Headset cameras	Thumb ring	Index ring
Select	same	same	Index pinch	Index pinch
Inspect / more	same	Index distal pad press-hold	Index middle pad tap	Index middle pad press-hold
Back	same	same	Middle distal pad tap	same
Menu / home	same	Middle middle pad tap	same	Middle distal pad stroke out
Dismiss / close	same	same	Middle middle pad stroke out	Middle middle pad tap
Scroll down	Index distal side stroke in	same	Index middle side stroke out	Index middle side stroke in
Scroll up	Index distal side stroke out	same	Index middle side stroke in	Index middle side stroke out
Next	same	Index distal side roll down	Index distal side roll down	same
Previous	same	Index distal side roll up	Index distal side roll up	same
Increase	Middle middle pad stroke out	Index distal pad stroke out	Hold middle pinch hold + wrist extend 30°	Hold c handshape + wrist extend 30°
Decrease	Middle middle pad stroke in	Index distal pad stroke in	Hold middle pinch hold + wrist flex 30°	Hold c handshape + wrist flex 30°
Zoom in	Middle distal pad stroke out	Index middle pad stroke in	Hold middle pinch hold + wrist ulnar 20°	Hold c handshape + wrist ulnar 20°
Zoom out	Middle distal pad stroke in	Index middle pad stroke out	Hold middle pinch hold + wrist radial 15°	Hold c handshape + wrist radial 15°
Changed	6/13	8/13	12/13	10/13

C Assumptions

- Frequencies (uses per minute): select 8, inspect 0.5, back 2, menu 0.5, dismiss 0.5, scroll 3 each way, page 1, value 0.5, zoom 0.3, focus left/right 4, up/down 2. Consequences of a misfire (wrong select = 1): inspect 0.3, back 0.7, menu 0.4, dismiss 0.8, scroll 0.2, page 0.3, value 0.5, zoom 0.2, focus 0.1.
- Scales: $H_0 = 2$ weighted false triggers per hour, $K_0 = 0.05$ wrong commands per use, $L_0 = 0.3$ s waiting per use.
- Features: contact weight, contact position and pressure 2 SD per unit; DIP and forearm ranges from study 1 (54° and 100°, 5th–95th percentile = 3.29 SD).
- Recognizer: speeds 0.67–1.5, amplitude 0.5–1.67, posture weight $\beta = 0.1$ on moving parts, moving part if RMS movement > 0.15 SD, drift $\nu = 0.75$ SD, forgiveness $c = 0.3$; soft edge 0.9–1.1; rest-like means less than twice everyday stillness (0.35 SD).
- Unobservable: more than half of the movement in forearm rotation, DIP or pressure.
- Search: 4,000 iterations per slot, 6 restarts for reported optima; 2,500 and 2 per uncertainty draw.

D Registration and method changes

The fifteen claims (`lib/vocab/predictions.ts`) were written after the literature pass and the plan, before any study 2 output, and their SHA-256 recorded (`docs/vocab-registration.md`); `scripts/vocab/check.mjs` fails if the file changes. The method then changed during development, after runs had printed claim outcomes: (1) the three phases of each everyday activity were joined; (2) the recognizer attends to the parts a gesture moves, with a posture term only

there (the first probe found no everyday match for any gesture); (3) tolerances became per gesture (one global tolerance makes small gestures match stillness and large ones never match); (4) speed and size were absorbed explicitly, with a residual tolerance (an isotropic tolerance made taps and holds lookalikes); (5) lens sensitivities scale each feature's share of the squared distance rather than its amplitude (cameras and the thumb ring had saturated every gesture); (6) unobservable gestures take their family's median rather than the global median (forearm rotation had looked free); (7) each body part is judged separately (the whole-vector tolerance let a held pinch with and without a wrist bend count as lookalikes); (8) the forgiveness was set to 0.3 by assumption (at 0.2 almost no gesture matched anything) after a calibration from contact-solver alternatives was rejected; (9) the recommended set is chosen by mean rank across draws rather than mean objective (objectives are not comparable across draws, and the mean let a few draws decide). Development runs printed 11, 7, 6 and 6 passing claims of 15 before changes 4–6, 7, 8 and 9 respectively. The predictions were never edited; the outcomes reported here are from the final method.

E Every single gesture

Gesture	Effort	Misfires per hour at $c = 0.15, 0.225, 0.3, 0.45, 0.6$					Closest
Thumb out	0.15	0	0	0	27	291	1.18
Open hand	0.16	0	0	0	0	3.0	1.84
Squeeze fingers together	0.20	0	0	0	0	0	never
C handshape	0.22	0	0	0	0	0	never
Thumb curl	0.23	0	0	0	32	501	1.07
Thumb across palm	0.24	0	0	0	12	309	1.32
Knuckle bend	0.24	0	0	0	3.0	85	1.34
Index distal side tap	0.26	0	0	0	1.5	89	1.41
Index distal pad tap	0.26	0	0	0	3.0	83	1.34
Hook	0.26	0	0	0	3.8	149	1.35
Index distal side press-hold	0.27	0	0	0	0	140	1.50
Index distal pad press-hold	0.28	0	0	0	3.8	140	1.26
O handshape	0.29	0	0	0	0	0.8	1.79
Index middle pad tap	0.30	0	0	0	0	6.8	1.59
Index middle side tap	0.30	0	0	0	0	159	1.53
Index pinch	0.30	0	0	0	3.0	278	1.15
Index pinch hold	0.31	0	0	0	5.3	321	1.25
Index distal pad stroke out	0.31	0	0	0	3.8	261	1.32
Index distal pad stroke in	0.31	0	0	0	2.3	264	1.35
Index distal side stroke out	0.31	0	0	0	2.3	245	1.33
Index distal side stroke in	0.31	0	0	0	1.5	240	1.27
Index middle pad press-hold	0.31	0	0	0	0	12	1.72
Index middle side press-hold	0.31	0	0	0	0.8	240	1.45
Index distal pad slide along	0.31	0	0	0	2.3	184	1.34
Index distal side slide along	0.32	0	0	0	1.5	168	1.30
Middle, ring + little curl	0.32	0	0	0	53	767	1.12
Index distal side roll up	0.33	0	0	0	2.3	247	1.29
Index distal side roll down	0.33	0	0	0	2.3	248	1.31
Middle distal pad tap	0.33	0	0	0	7.6	137	1.34
Index distal side slide across	0.33	0	0	0	1.5	177	1.30
Pinch press	0.33	0	0	0	5.3	321	1.25
Index middle side roll up	0.34	0	0	0	3.0	421	1.41
Middle middle pad tap	0.34	0	0	0	0	111	1.60
Index middle side roll down	0.34	0	0	0	2.3	406	1.44
Middle distal pad press-hold	0.34	0	0	0	4.5	220	1.16
Index middle side slide across	0.34	0	0	0	2.3	309	1.42
Middle pinch	0.34	0	0	0	9.1	254	1.38
Close hand	0.35	0	0	0	0	0.8	1.78
Middle middle pad press-hold	0.35	0	0	0	0	217	1.51
Middle pinch hold	0.35	0	0	0	8.3	332	1.29
Index middle pad stroke out	0.36	0	0	0	0	204	1.59
Index middle pad stroke in	0.36	0	0	0	0	164	1.52
Ring + little curl	0.36	0	0	0	77	853	1.09
Middle distal pad stroke out	0.37	0	0	0	9.9	372	1.19
Middle distal pad stroke in	0.37	0	0	0	7.6	362	1.23

Gesture	Effort	Misfires per hour at $c = 0.15, 0.225, 0.3, 0.45, 0.6$					Closest
Index middle pad slide along	0.38	0	0	0	0	113	1.60
5 handshape	0.38	0	0	0	11	413	1.33
Index middle side stroke out	0.39	0	0	0	0	464	1.53
Index middle side stroke in	0.39	0	0	0	3.0	481	1.42
Middle distal pad slide along	0.39	0	0	0	4.5	274	1.16
Ring distal pad tap	0.39	0	0	0	0.8	32	1.45
Index middle side slide along	0.40	0	0	0	1.5	364	1.44
Ring distal pad press-hold	0.41	0	0	0	1.5	89	1.33
A handshape	0.41	0	0	0	0	35	1.64
Ring pinch	0.41	0	0	0	0.8	39	1.48
Little distal pad tap	0.41	0	0	0	0.8	214	1.44
Middle middle pad stroke out	0.41	0	0	0	0.8	487	1.45
Middle middle pad stroke in	0.41	0	0	0	0	476	1.54
Little pinch	0.41	0	0	0	1.5	125	1.25
Ring pinch hold	0.41	0	0	0	0.8	39	1.36
Little pinch hold	0.41	0	0	0	3.0	102	1.37
Little distal pad press-hold	0.42	0	0	0	3.0	278	1.46
Ring middle pad tap	0.42	0	0	0	0	141	1.45
B handshape	0.42	0	0	0	0	118	1.52
Middle middle pad slide along	0.43	0	0	0	0	349	1.58
Ring middle pad press-hold	0.44	0	0	0	0	289	1.51
S handshape	0.44	0	0	0	0	117	1.49
Little middle pad tap	0.45	0	0	0	0.8	265	1.47
Little middle pad press-hold	0.45	0	0	0	0	398	1.53
L handshape	0.46	0	0	0	0	20	1.73
Fist squeeze	0.47	0	0	0	0	70	1.44
Little distal pad slide along	0.48	0	0	0	3.0	326	1.44
1 handshape	0.49	0	0	0	0	36	1.75
Y handshape	0.49	0	0	0	0	290	1.73
Ring distal pad slide along	0.49	0	0	0	1.5	96	1.38
Ring middle pad slide along	0.49	0	0	0	0.8	416	1.48
I handshape	0.50	0	0	0	0	71	1.71
3 handshape	0.50	0	0	0	0	133	1.77
W handshape	0.51	0	0	0	0	116	1.77
V handshape	0.52	0	0	0	0	168	1.73
Horns	0.53	0	0	0	0	69	1.77
Spread fingers	0.22	0	0	0.8	89	479	1.00
Index curl	0.26	0	0	0.8	216	1180	0.89
Index + middle curl	0.32	0	0	0.8	71	802	0.94
Little curl	0.35	0	0	1.5	158	1254	0.89
Ring curl	0.41	0	0	1.5	184	1358	0.97
Little lift	0.25	0	0	4.5	149	564	0.79
Ring lift	0.37	0	0	4.5	122	495	0.86
Middle lift	0.32	0	0	6.8	167	690	0.89
Index lift	0.18	0	0	7.6	189	827	0.87
Middle curl	0.35	0	0	7.6	239	1411	0.80
Wrist ulnar 20°	0.17	0	3.0	11	150	592	0.61
Wrist radial 15°	0.18	0	0	11	101	397	0.83
Forearm pronate 45°	0.12	1762	1762	1762	1762	1762	0.00
Forearm supinate 45°	0.13	1762	1762	1762	1762	1762	0.00
Wrist extend 30°	0.11	0	0.8	11	95	389	0.76
Wrist flex 30°	0.12	0	0.8	11	96	332	0.71

* Mostly unobservable in the everyday data; the optimiser uses an estimate.

F Reproducibility

- `.venv-analysis/bin/python-Ianalysis/everyday_stream.py` builds the everyday stream from `KIN_MUS_UJI.mat`.
- `node scripts/vocab/run.mjs` computes everything (heavy stages cached in `analysis/output/vocab/`); `--quick` for development.
- `node scripts/vocab/paper-tables.mjs` and `analysis/vocab_figures.py` render the tables and figures; `node scripts/vocab/check.mjs` checks the registration, invariants and outputs.

The demonstration at <https://ethanlambeth.com/hand-input/> plays every gesture, replays everyday movement against the vocabulary and runs the optimiser in the browser.